TestPilot: An Optimized Framework for Scalable, Cross-Browser Automated Testing

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Abstract— TestPilot is an advanced automation testing framework leveraging Playwright and JavaScript to address limitations in manual testing and existing tools. It ensures cross-browser compatibility, dynamic test execution, and smart bug detection, enhancing software quality assurance. With an intuitive interface and seamless integration into CI/CD pipelines, TestPilot optimizes testing efficiency and scalability for modern applications. The Agile methodology ensures iterative development and user-centric design. Performance results demonstrate improved test execution speed, reduced human errors, and enhanced ability to manage large-scale dynamic test suites, making TestPilot a robust solution for achieving reliable and high-performing software products.

Keywords—Automated Testing, Playwright, Cross-Browser Testing, CI/CD Integration, Dynamic Bug Detection, Web Application Testing.

I. INTRODUCTION

Delivering high-quality applications becomes a necessity in today's fast-paced software development scenario to meet user expectations as well as remain competitive in the market. Automation testing plays a vital role when it comes to ensuring that products are bug-free, efficient, and reliable. A project that uses Playwright and JavaScript for automating the testing process is here discussed, which addresses the inefficiencies present in the manual method of testing. TestPilot is set to revolutionize the paradigm of software quality assurance using speed, accuracy, and scalability through modern frameworks and intelligent testing techniques.

The development of TestPilot is inspired by the need to overcome the limitations of ordinary testing tools and methods that have traditionally been used; the main drawbacks of any form of manual testing lies in the time-consuming natures of these tests combined with their propensity to having errors, and existing auto-tools struggle with performance, scale, and dynamic web applications as well.

A. Relevance Of The Project

Software testing is a critical phase in the software development lifecycle (SDLC), ensuring the delivery of reliable and high-performing applications. However, manual testing methods are inefficient for large-scale projects, and existing tools like Selenium often require steep learning curves and extensive configuration. Modern software applications require faster release cycles, demanding more efficient, automated, and scalable testing solutions.

Software testing forms that critical stage in the software lifecycle which must provide reliable and quality performing applications. Nevertheless, any manual testing technique of today's large-scale projects aren't too efficient; already developed tools like Selenium ask much in terms of steep learning curves besides much configuration and a very critical area of which applications under modern software requirements require speedier release cycles.

The current method of software testing is very challenging. It involves manual testing, which is slow, tedious, and prone to human error. This makes it ineffective for modern apps that require speedy updates. Human effort can't support agile development cycles. Some automation tools offered commercially are somewhat of disadvantages, being Selenium, to name just an example-they can get complicated, take up hours to know how to master, and sometimes the output of your web applications with them doesn't really fly. Scalability is another big concern in this area as most currently implemented solutions have poor tests with scale tests, making them bad choices for more complex flows in CI/CDs.

B. Software Development Methodology

The TestPilot project is Agile Software Development, which is best suited for projects that require frequent iterations and continuous improvement. Some of the main characteristics of this methodology are:

- Iterative Development: The entire development cycle is divided into sprints wherein a function version of the framework is evolved at each step.
- Collaboration: Developers, testers, and stakeholders need to be in constant collaboration for the tool to meet the needs of end-users.
- Continuous Integration: The code and changes are continuously integrated so that functionality does not break with respect to the evolving framework.
- User-centric design: With usability in focus, TestPilot should be easy to use by all kinds of users: from a tester to developers.

The design strategy is flexible to answer user feedback and project requirements, which will ensure that the end result is functional and effective.

Agile

Agile is a method in which a team can manage a project through dividing the project with continuous stakeholders' involvement, iterative improvement and build-in-cycle at each phase of project development. It promotes continuous testing and continued development throughout the lifecycle of the software development project.

Fig 1.1 Agile Methodology

II. LITERATURE SURVEY

This chapter discusses the existing techniques in automated testing, analyzes reference papers, highlights the gaps identified in current solutions, and presents inferences derived from the survey.

A. Overview of Existing Work

Automated testing is an important part of software quality. The most popular tools for E2E testing are traditional ones like Selenium, but they have several drawbacks, such as long execution times and a steep learning curve. Modern frameworks, such as Playwright and Cypress, try to tackle some of the issues but still struggle with scalability and bug detection. Current methods include these techniques:

- Input Preprocessing: Tools prepare the input for dynamic user interfaces by structuring it and setting up testing environments [1].
- Cross-Browser Testing: Many tools automate testing across browsers, but achieving consistent results in dynamic content remains challenging.
- Test Case Optimization: Modern frameworks employ dynamic test case prioritization based on resource availability, reducing execution time and improving efficiency [2].

These techniques offer minor improvement but have a problem in large-scale complex testing needs and therefore the need for smarter, scalable solutions.

B. Reference Paper

Evaluating the existing work will be important to understand advancement and limitation in automated testing. This section presents key research papers from the literature survey, with focus on tools, methodologies, and their effects on software quality, cost, and time-to-market

Comparison with other papers

The strengths and weaknesses of current automated testing tec hniques come out as the methodologies compared in different papers.

1) Preprocessing and Dynamic Testing

The need for test case preprocessing: This compatibility requirement exists with complex user interfaces and is the reason why automated testing is necessary. Traditional tools, like Selenium, require heavy manual setup, which leads to error-prone and time-consuming processes, especially when dealing with large projects. Even though Selenium comes in various configurations, it does not work well with modern, state-dependent applications. Selenium's static nature restricts its effectiveness with dynamic content, resulting in inconsistent outcomes in various environments [1].

Cypress enhances Selenium by making preprocessing easier and automating configuration for testing dynamic web pages. However, it is still demanding in terms of effort for state-dependent applications and complex interfaces without modularity for comprehensive dynamic content testing. Even with its progress, Cypress lags behind newer tools such as Playwright. Modular, reusable components help the playwright shine at preprocessing and dynamic testing. It simplifies test setup and increases flexibility, thus allowing efficient dynamic script writing. Its architecture supports the testing of complex, state-dependent applications to ensure greater compatibility with modern web technologies. Its advanced capabilities support dynamic content, reduce the effort of the creation of test cases, and yield reliable results with less manually setup compared to Selenium and Cypress [1].

2) Execution Time Optimization

Another parameter to judge automated testing tools is execution time. Selenium is prone to performance issues as it carries out test executions in sequential order and requires page reloads, thereby increasing the runtime [2].

Cypress improves the performance of execution by loading DOM in real-time, thereby reducing test delays. On the contrary, Playwright leads through parallel execution, where a number of tests can run in parallel across browsers and platforms, which reduces runtime significantly. This feature along with headless browser support makes for quicker feedback and efficient usage of resources. This is to say that the ability to run multiple tests simultaneously puts playwright at a competitive edge with big projects and large test suites in comparison with Selenium and Cypress.

3) CI/CD Pipeline Integration

Modern software development utilizes CI/CD pipelines to rapidly deliver high-quality software. Based on reviewed studies, Selenium, being popular in its use cases, lags in terms of the integration with CI/CD pipelines because it relies mostly on manual configurations and very complex dependency management [3].

Tools such as Cypress integrate better but still demand further pipeline configuration. Meanwhile, Playwright simplifies CI/CD natively because it supports Jenkins and GitHub Actions directly, so integrating automated testing is pretty straightforward and cuts down the time spent on configuration while frequently testing. Playwright outshines Selenium and Cypress when it comes to CI/CD automation, which is perfect for agile environments where continuous testing is a necessity [1].

4) Cost, Quality, Scalability

Automated testing tools greatly impact the cost, quality, and scalability of software projects. Selenium is popular but is not suited for large projects. It tends to execute sequentially, slowing down performance as test cases increase and leading to longer runtimes and increased resource usage. With high testing, selenium requires more resources hence increasing the costs and time-to-market. Furthermore, it faces a great challenge in handling complex dynamic content hence resulting in lower quality testing in some environments [1].

Because real time DOM reloads and also waits automatically, Cypress executes testing much faster than the equivalent in Selenium for relatively small to medium applications; however, it does falter at scale as large and complex projects are designed. In addition, as more large projects require support multi browser testing, its use gets restricted. Even though it increases the efficiency of testing, it is much more expensive for integration and maintenance in large-scale environments because it requires manual configuration in complex workflows [2].

Strong cross-browser capabilities and parallel execution address the scalability issues of Selenium. It then comes in handy in extensive suites of tests, an inevitable requirement in large-scale tests. Tests can then execute in parallel across all chosen browsers, thereby enhancing speed as well as reducing the actual amount of test time plus also reducing the cost. The dynamic bug detection and reporting of errors improve the testing quality by quickly identifying the defects. Scalability, performance, and cost effectiveness make Playwright an excellent choice for organizations looking forward to scaling testing while still maintaining high standards [4].

Feature	Selenium	Cypress	Playwright
Preprocessing	Manual Configuration	Semi-Automated	Fully Modular Setup
Execution Time	Longer	Moderate	Optimized (Parallel)
Dynamic Content Testing	Limited Support	Moderate Support	Advanced Support
CI/CD Integration	Complex	Moderate	Seamless Integration
Scalability	Performance degradation	Moderate scalability	Efficient and scalable

Table 2.1 Comparative Analysis of Automated Testing Tools

C. Gaps Identified

This practice involves an automated testing tool, although there are gaps in its method.

- High Learning Curve: Tools like Selenium demand extensive configuration and expertise, making them challenging for beginners [2].
- Performance Bottlenecks: Execution times are considerably increased, especially in headless modes or with large test suites [1].
- Limited Dynamic Testing: Traditional tools cannot handle state-dependent dynamic web applications, leading to incongruous testing results [3].
- Insufficient Integration: Existing solutions struggle with integration into modern CI/CD pipelines, reducing their effectiveness in agile and DevOps environments [4].

These gaps represent a demand for an intuitive, scalable, and strong testing framework that addresses:

- Decreases learning curve with fewer workflows:
- Improved performance with parallel execution.
- Manages dynamic web content effectively.

D. Inferences from Survey

The Key inferences from the literature survey include:

- Performance Optimization: Playwright outperforms compared to Selenium and Cypress in its execution time as well as resource usage. Parallel testing helps in saving much of the time overhead [1].
- Dynamic Content Management: Classic tooling struggles with state-dependent applications. Playwright is doing a great job with supporting modern web interactions with high efficiency [2].
- Ease of Use: Selenium's learning curve and manual setup are barriers to adoption. Playwright simplifies this with modular architecture and reusable components, cutting development time [3].
- Scalability: Large-scale testing frameworks often experience degradation in performance. Playwright efficiently handles large test suites through cross-browser support and parallel execution [4].
- CI/CD : Very efficient for continuous testing as it integrates well with tools, such as Jenkins and GitHub Actions [1].

Chapter 2 reviews the automated testing tools: Selenium, Cypress, and Playwright. The superior performance, scalability, dynamic content testing, and CI/CD integration of Playwright are mentioned. Gaps in these tools include steep learning curves, performance issues, and limited CI/CD support. Intuitive frameworks like Playwright are in demand to meet the current testing challenges.

III. SYSTEM DESIGN

The current chapter elaborates on the TestPilot framework in terms of system design, algorithms, data flow, and technologies implemented. TestPilot is an automated testing solution based on the Playwright framework that allows conducting efficient, dynamic E2E web application testing. This system is developed to support even the most complex, dynamic web content and seamlessly integrates with CI/CD pipelines. The chapter is divided into sections which discuss different features of the system design

A. System Architecture

The TestPilot framework is based on modular and efficient architecture, using major components like Playwright for execution, bug detection, and reporting. That way, it can be executed across multiple browsers, with management of dynamic web apps and easy integration into CI/CD workflows.

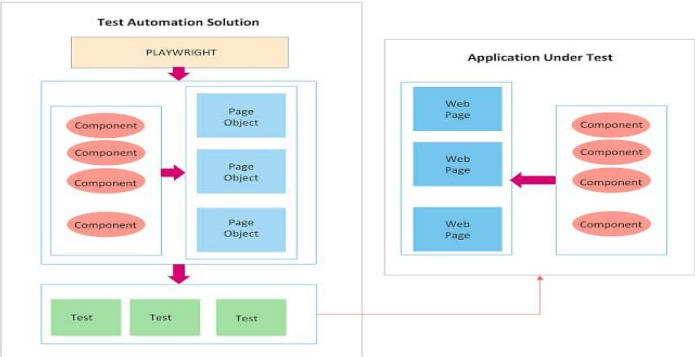

Fig 3.1 System Architecture

The TestPilot framework separates the Test Automation Solution from the AUT. It is a Core Tool in the form of Playwright, which acts as a tool to automate the browsers. Modular Design: it is an organized way in which tests can be created as modules composed of certain actions or features, thereby ensuring proper maintainability and reusability. Components map to Page Objects that reflect the structure and behavior of web pages. These Page Objects then interact with the Application Under Test through web pages for navigation, element interaction, and assertions.

The target system consists of web pages and components. The work of Playwright is done to simulate user interactions by testing the functionality. Different Page Objects are defined with the buttons and forms; thus, tests run on these components either in series or in parallel. Thereafter, the results are collected and analyzed. This flow systematically validates functionality, finds bugs and is highly efficient in recording outcome; it also emphasizes a modulated approach to test automation by maximizing scalability and minimizing maintenance.

Block Diagram of the Proposed System

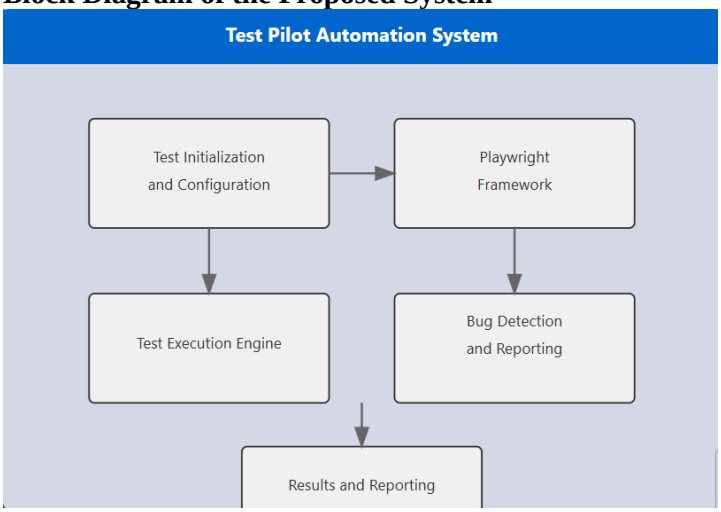

Fig 3.2 Block Diagram of the Proposed System

1) **Initialization and Configuration of Tests:** It sets up the test environment by preparing configurations for any necessary setup, such as initialization of browser instances, and loading configurations like authentication tokens, base URLs, etc.
2) **Playwright Framework:** The center testing engine interacting with the browses. It drives and simulates user interactions and does assertions. This actually executes the tests and captures all logic-related data in terms of its ability to interact with the web elements.
3) **Bug detection and reporting:** Detect bugs at runtime. The bug is logged. Provides a detailed report of the test run.
4) **Results and Reporting:** Collects, organizes, and presents test results in a user-friendly format (logs, screenshots, videos, etc.), often integrated into bug tracking tools or CI/CD pipelines.

Overview of Playwright Framework

Playwright is one powerful and robust automation framework to simplify browser automation. It is developed by Microsoft and supports multiple programming languages. It can be applied to simplify browser automation and increase the quality of web application testing. It allows teams to run their tests in both headless as well as headful modes. As a result of the Debugger mode being accessible, software engineers can make use of this capability to really analyze and debug problems and anomalies to ensure that the app is free from errors. Additionally, besides other many features, it is believed that Playwright is one of the first options for modern browser automation.

Page Object Model Pattern using JavaScript with Playwright

A widely used design pattern in automated testing is called Page Object Model, or POM. It is an abstraction of a part of a web application. It means that a web application might have a home page, listings page, checkout page, and contact page. All these can be represented using page object models. Page objects are a classical example of encapsulation – it hides the details with methods and variables in order not to clutter the tests.

The core concept of a Page Object Models pattern, therefore is to create a class file for each and every webpage or component in a web application. Every such class file hosts only the corresponding web page elements; these classes encapsulate their UI elements along with associated functionalities. Each page object is an abstraction of a web application page and gives methods for the interaction with the web elements that are being shown on that specific web page, such as buttons, checkboxes, or text fields-for example, password field. It shall certainly contain a page object for a section list page with cards like this, as each online service displays cards.

Similarly, an online masthead might also contain a header page object for that. Similarly, if we jump to another page, the page object of the first page should return another page object for the new page. Generally, page object operations should return primitive types (strings, dates) or other page objects.

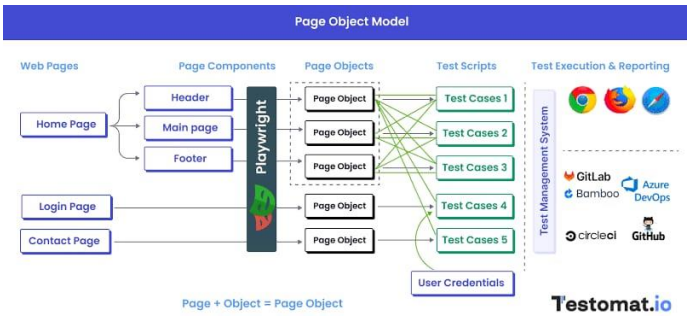

Fig 3.3 Page Object Model: JavaScript with playwright

Although it's called the Page Object, the objects should not typically be instantiated per page but rather for meaningful parts on a page. A rule of thumb is to model the structure in the page that makes sense in order to improve our testing. Modelling the application's UI into a dedicated object called a Page Object assists teams not only to get code duplication low but to improve the test framework in general.
The major benefits that playwright offers over older tools like Selenium include:

- **Cross-Browser Support:** Playwright can run tests on Chromium, Firefox, and WebKit, ensuring that web applications are tested on the most popular browsers all with one unified API.
- **Parallel Execution:** Playwright allows running tests in parallel on multiple browsers. This saves a lot of time for test execution and optimizes resources.
- **Headless Mode**: It makes the Playwright run tests in a headless mode and does not open any graphical interface.
- It supports modern web applications. It can support dynamic, often changing content modern web applications based on user actions. This is particularly essential in testing Single Page Applications and applications with rich user interfaces.

These Playwright features are leveraged by TestPilot to create a more efficient and dynamic testing process that enables tests to be run fast and reliably in different environments.

A. Proposed Algorithm

The TestPilot framework represents the process of executing automated tests, which begins from initialization of the environment to dynamic generation and execution of test cases. In fact, the described steps explain how each stage of the test execution would be orchestrated.

Environment Initialization

The Environment Initialization process is a critical step as it prepares the entire test environment before the execution begins. The algorithm will initialize browser instances and loads configurations, ensuring that all necessary dependencies are met before tests run.

1) **Initialization of start environment:** Input test configuration such as which browsers to use, base URL, environment variables.
2) **Creates Playwright Browsers:** The code initializes the Playwright browser setup by creating instances of different browsers, including Chromium and WebKit, with respective given settings. It activates a headless mode on specified browsers for faster executions only when required. It separately configures each of their instances with isolated contexts; each of them runs a completely detached test to avoid dependency so that the test would surely run efficiently and dependently every time.
3) **Configure Browser Context:** For every browser, so the isolated contexts are provided for sessions and data regarding tests. Start pages to ensure they have a fresh, clean state for every test.
4) **Setting up test configurations:** Inject necessary configurations such as the base URLs of your web application, environment variables such as API keys and authentication tokens, session-specific data such as login credentials and user details, among others.
5) **Check Environment Setup:** Ensure all configurations loaded and no missing dependencies are there.
6) **End Environment Initialization:** The output is a clean, isolated and fully configured environment ready for test execution.

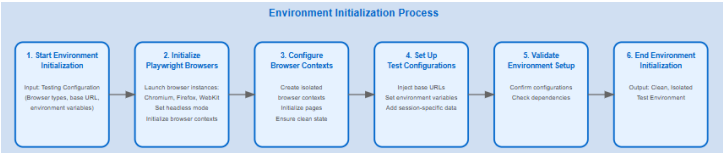

Fig 3.4 Environment Initialization

This ensures a clean, isolated, and well-defined environment for each test run, which will allow reproducibility and consistency in test execution.

Dynamic Test Case Generation

This will optimize, with regard to certain criteria in the selection of test cases, the test suite under the algorithm of Dynamic Test Case Generation. What that essentially means is the unnecessary overhead of only the pertinent tests being executed.

1) **Dynamically Generate the Test Case:** The application functionality, current state of the application, test suite configuration, and priority setting are taken as input.
2) **Identification of Testing Requirements:** Testing requirements can be identified through how the application works, new additions it has recently, modified code, hot spots that usually error out, and how to handle its login system or payment gateway.
3) **Dynamic Test Case Generation:** For each requirement identified, automatically generate relevant test cases:If new functionality is being tested, generate functional tests for new features. If bug fixes are being validated, generate regression tests for affected components.
4) **Testcase Prioritization:** Based on predefined priority rules (example frequently used functionalities and high-risk components have a high priority to execute. Perform tests in ordered lists prioritized with highest prioritized one first.
5) **Test Case Optimization:** Optimize test execution by prioritizing simpler tests to run first, reducing overall execution time. When multiple browsers are available, assign fewer complex tests to faster browsers to enable parallel execution and improve efficiency.
6) **Dynamic Test Case Generation with Priority:** The output is the optimized set of test cases that are dynamically generated and prioritized for the current state of the application.

Cross-Browser Execution

Cross-browser execution is important because it enables cross-browser testing of the consistency of a web application in multiple browsers. Because of this cross-browser testing capability, Playwright is helping TestPilot run the same test suite on multiple browsers for compatibility and uniformity.

1) **Start Cross-Browser Execution:** The Input is
   a) Test suite (set of test cases).
   b) Browser selection configuration, which will be either Chromium, Firefox, or WebKit.

2) **Initialize Playwright Browser Instances**: Launch the browser instances of Chromium, Firefox, and WebKit.
   a) **Step 1:** For every browser in the configuration. Self-start an independent Playwright browser session. Develop various scenarios for every browser instance in order to isolate and test it accordingly.
   b) **Step 2:** Each browser instance should be initialized in parallel and prepare all the necessary browser contexts for running tests.
3) **Run parallel tests on multiple browsers**
   a) **Step 1:** For each browser instance, Chromium, Firefox, and WebKit, run the same test suite in parallel. Playwright runs tests for each browser in parallel, significantly reducing total execution time.
   b) **Step 2:** Test the same tests on all the browsers to ensure consistency and comprehensive coverage. The test suite might have different kinds of tests: functional, regression, or compatibility.
4) **Fetch Outputs from All Browsers**
   a) **Step 1:** Collect the following for each browser instance while running the tests
      i) Test Results: Pass/Fail status for each test case.
      ii) Logs: Capture browser console logs, network requests, and errors.
      iii) Screenshots: Capture screenshots of the page for the failed test cases.
      iv) Videos: The test runs can be recorded into a video for fine debugging insights.
   b) **Step 2:** Collect the results in a structured format, such as logs and reports, for comparison and analysis.
5) **Interpret results and note inconsistencies**
   a) **Step 1:** After testing all the browsers, results are compared. If there is a behavioral mismatch (for example, if one browser fails to pass when others pass, flag it as browser-specific). For example, if a test passes on Chromium but fails on WebKit, then there is probably a compatibility issue with WebKit
   b) **Step 2:** Record details of discrepancies, including which type of failure, which version of browser, and against which specific test case where the failure occurred. Record browser-specific bugs or inconsistencies for later review and debugging.
6) **Create Cross-Browser Test Report**
   a) **Step 1:** Collect an accurate document that contains:
      i) Results for each of the browsers.
      ii) A summary of browser-specific issues or discrepancies detected during testing.
      iii) Comparison of test behavior across different browsers.
   b) **Step 2:** Save the report and send it to the development or QA team for further analysis and bug fixing.
7) **End Cross-Browser Execution Output**: Results of the test performed and differences found on different browsers. Browser-specific issues flagged for debugging. This will ensure that the web application works in all environments and detects browser-specific bugs or inconsistencies early in the testing process.

Dynamic Bug Detection

The Dynamic Bug Detection process runs alongside test execution, monitoring for failures or issues throughout the test lifecycle.

1) **Enable Dynamic Bug Detection Test environment, browser, and results.**
2) **Monitor Browser Console Logs:** While testing:
   a) Capture browser console logs (JavaScript errors, network requests, warnings).
   b) Find issues such as JavaScript errors, failed network requests, or rendering problems.
3) **Take Screenshots and Videos If a failure occurs:**
   a) Capture a screenshot of the current page.
   b) Optionally, record a video of the test for debugging.
   c) Record its timestamp, error message, with the relevant details of the browser log.
4) **Real-Time Bug Reporting When a bug is found:**
   a) Log the bug with details like error type, browser version, and session data.
   b) Create a bug report with all relevant details (logs, screenshots, videos).
   c) Integrate with JIRA or similar tools to automatically create a bug ticket.
5) **End Dynamic Bug Detection**: Real-time bug reports with screenshots, logs, and videos for analysis. This captures failures very quickly and logs all relevant data for post-test review.

B. Data Flow & Design

The Data Flow and Design section of TestPilot outlines the test execution processes, from environment setup to report generation. It ensures smooth operation by efficiently managing browser initialization, test case execution, bug detection, and result collection.

System Flowchart

The System Flowchart below gives the sequence of operations in the TestPilot framework to ensure that the proper steps are followed in a particular order for an efficient automated testing process.

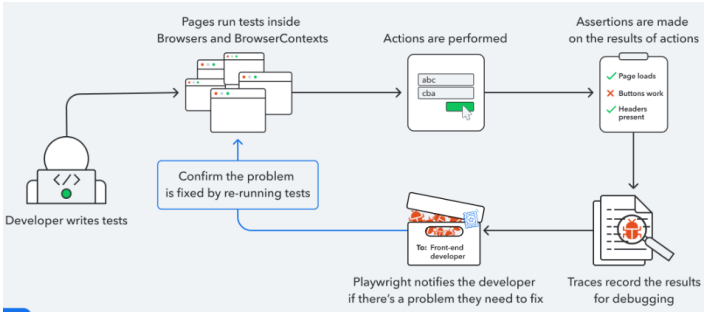

Fig 3.5 System Flowchart

Browsers supports using web browsers like Chrome, Firefox, Edge, and Safari to test. Browser contexts are allowed to run tests concurrently in isolated browser sessions.
Pages are individual tabs or windows in a BrowserContext, used to load documents, navigate links, and interact with content.

Tests execute actions and throw assertions if the results do not match. If an assertion fails, it means that the test does not match the value expected.

Traces capture information at the time of testing and can be used for debugging purposes. Trace data points out where the problem lies during a test that fails.

Locators use a DOM selector to find page elements. Playwright can check these elements with assertions or interact through actions.

This chapter explains the TestPilot framework in terms of its design, algorithms, data flow, and technologies. It is an automated testing solution that is based on Playwright and supports dynamic E2E web application testing across browsers. It integrates well with CI/CD pipelines and handles complex web content efficiently. The system is designed using a modular architecture. It utilizes Playwright as the primary tool for browser automation and reporting. It utilizes the Page Object Model for better maintainability and reusability. TestPilot utilizes cross-browser support and parallel execution features of Playwright to enhance test performance.

Details include environment setup, test case generation, cross-browser execution, and bug detection. The algorithm is based on isolating browser contexts, optimizing tests by complexity, and detecting issues in browsers. Real-time dynamic bug detection captures console logs, screenshots, and videos to produce an accurate report. Finally, the chapter ends with discussions of data flow and design within the system, showing how the framework handles browser setup, test running, and result gathering for seamless automated testing.

IV. IMPLEMENTATION

This chapter describes TestPilot implementation for automating web application testing with Playwright. It includes information on algorithm details, the experimental setup, comparison algorithms, performance metrics, and front-end and back-end implementation. The chapter includes test case design and execution details, such as dynamic bug detection.

A. Algorithm Details

It has the advantage of structured algorithms to execute automated tests in an efficient manner; they optimize the execution time, reduce the time consumed, and improve bug detection in browsers.

1) **Environment Initialization Algorithm**:

The algorithm creates a testing environment with Playwright browser instances initiated with configuration settings: Chromium, Firefox, WebKit. In isolation, browsers are used to prevent sharing of session data. Configuration values such as the base URLs, API keys, and authentication tokens are set in environment before running tests. All this is done for clean reproducible setup during each run.

2) **Dynamic Test Case Generation Algorithm**:

This algorithm prioritizes tests based on the application's state. It analyzes functionality to identify features needing testing, like new features or critical bug fixes. Tests are ordered by risk and usage; the most important feature will be tested first, so the test suite first runs the high-priority tests to reduce overall testing time.

3) **Cross-Browser Execution Algorithm**:

The system initializes multiple Playwright browser instances for Chromium, Firefox, and WebKit to run tests in parallel, reducing execution time.Results of testing on all browsers are collected, and behavior discrepancies are noted for investigation. This ensures that the application functions uniformly across major browsers.

4) **Dynamic Bug Detection Algorithm**:

During testing, the system monitors for failures and captures logs, which include console logs, network issues, and JavaScript exceptions. The playwright captures screen shots or videos of the failure for debugging. An algorithm dynamically detects and logs bugs for quick resolution.

B. Experimental Setup

To analyze TestPilot performance, set up particular hardware and software that should work compatibly and efficiently for automation testing across several browsers.

Hardware Requirements

The key hardware components in TestPilot's setup are these:

- **Processor:** With multiple cores such as Intel Core i7 or AMD Ryzen 7, it reduces processing time in case of parallel execution on multiple browsers.

- **RAM:** A minimum of 16GB of RAM is mandatory to run multiple browser instances and resource-heavy tests.

- **Disk Space:** Free space on disk needs to be at least 100 GB for test logs, screenshots, videos, and other artifacts.

- **Network:** A stable high-speed internet connection is required for smooth downloading of dependencies and remote test execution in the cloud.

Software Requirements

The TestPilot project, which uses Playwright and JavaScript for test automation, requires certain software components to be operational. Following is a breakdown of the software requirements for TestPilot:

- **OS**: TestPilot works with several major operating systems, making it flexible for development.
  a) Windows 10/11: It ensures compatability in Windows, which most developers and test engineers use.
  b) macOS: This will be helpful to teams working with the assistance of Apple hardware as the tool will be available within macOS environments.
  c) Linux, such as Ubuntu, since TestPilot is tested for Linux, especially Ubuntu, which is commonly used for server and cloud testing.

- **Node.js:** TestPilot uses Playwright, running test automation scripts on it, but such scripts are technically Node.js apps because to run those files, JavaScript code

should have been issued by the application. Node.js 12.x+: Compatible and high performance for running Playwright with modern JavaScript

- **Playwright:** Playwright is an automation framework for web browser interaction. It supports cross-browser testing on Chromium, Firefox, and WebKit to ensure web applications function consistently in different environments. Ensure latest stable Playwright install for compatibility with the given browsers and for access to additional features such as headless testing, parallel execution and more advanced debugging.
- **Browsers (Chromium, Firefox, WebKit)**: TestPilot allows auto-testing for three most important browsers.
  a) Chromium (Chrome, Edge): Extremely popular with auto-testing.
  b) Firefox: A popular browser known for its developer tools and open-source nature.
  c) WebKit: Essential for testing web apps on Safari or Apple devices. The playwright runs tests in multiple browsers at once, ensuring web app compatibility and performance across major platforms.

- **Version Control (Git):** This version provides enablement of change tracking. Git enables change tracking that provides branch management for TestPilot scripts, thereby it allows them to share their changes efficiently among the team with no loss in error in testing.

- **CI/CD Tools:** TestPilot integrates with CI/CD tools to test automatically within the development pipeline. The supported tools are:
  a) Jenkins: Most famous for automating testing and building in CI/CD pipelines.
  b) GitHub Actions: A CI/CD tool in GitHub for automating testing workflows triggered by changes in code or pull requests.
  c) GitLab CI: A well-known CI/CD tool, inside GitLab, for auto-testing and deployment.

- **Bug Tracking (JIRA or Similar):** TestPilot integrates with JIRA or similar tools for the purpose of logging and defect management when defects are identified at testing time.

C. Front-End and Back-End Implementation

The TestPilot framework's execution is separated into two fundamental components: the front-end (client interface) and the back-end (the motor that handles test execution, setup, and information handling). Each component is outlined to guarantee that the device is both simple to utilize and exceedingly productive in executing computerized tests.

Front-End Implementation

The front-end of TestPilot is created utilizing ReactJS, a well known JavaScript library for building client interfacing. It gives an intelligently and responsive UI that permits clients to effortlessly associated with the instrument and control the test computerization prepare.

The fundamental highlights of the front-end include:

- Dashboard: The dashboard gives clients with an outline of the current test runs, counting test status, test execution time, and comes about. It is outlined to provide analyzers speedy get to to the most critical information.
- Configuration Board: Clients can arrange test settings through a user-friendly interface. The board permits analyzers to select which browsers to test against, characterize test parameters, set environment factors (e.g., API keys), and design the test environment for execution.
- Test Comes about See: After tests have been executed, the comes about are shown in a clear and easy-to-understand organize. Each test case is appeared with its execution status (pass/fail), and point by point logs are accessible to offer assistance distinguish issues. Screenshots and recordings are given for fizzled tests to help with debugging.
- Error Announcing: When a bug is identified, an mistake report is naturally created with all pertinent data (logs, screenshots, etc.). Clients can see these reports straightforwardly in the front-end UI. The front-end permits clients to control and screen the whole testing handle, guaranteeing that they can make educated choices approximately the quality of the application beneath test.

Back-End Implementation

The back-end of TestPilot is dependable for executing test scripts, overseeing assets, and taking care of the organization of Writer test mechanization. The back-end is built utilizing Node.js and communicates straightforwardly with the Writer mechanization device to connected with web browsers and execute the test cases.

The key duties of the back-end include:

- Test Execution Motor: The center of the back-end is the test execution motor, which oversees the stream of test execution over distinctive browsers. The motor starts browser occurrences (Chromium, Firefox, WebKit), plans test scripts, and dispatches the tests. It too collects comes about from each test run and passes them to the front-end.
- Configuration Administration: The back-end stores all setup subtle elements, such as the test environment settings, test suites, and chosen browsers. It makes beyond any doubt that all environment factors (e.g., URLs, qualifications) are accurately set some time recently executing any tests.
- Test Information Administration: The back-end handles test information, such as logs, screenshots, recordings, and mistake reports. It forms and stores comes about in a organized arrange for simple recovery and analysis.
- Integration with CI/CD: The back-end coordinating with CI/CD frameworks (e.g., Jenkins, GitHub Activities) to permit robotized test execution as portion of the program conveyance pipeline. It gives API endpoints for activating test runs, recovering test comes about, and collaboration with the CI/CD workflow. Together, the front-end and back-end work in pair to give a consistent encounter for mechanizing and overseeing test execution.

D. Test Case Design and Execution

Execution Test case plan is significant for guaranteeing that the mechanized tests viably cover the key perspectives of the web application. TestPilot employments a combination of inactive

and energetic test case era to guarantee that tests are not as it were comprehensive but moreover productive. Underneath are the key steps in the test case plan and execution process.

Test Suite Overview

The TestPilot test suite is composed of different categories of tests, each planned to guarantee distinctive perspectives of the application are tried thoroughly:

- Functional Tests: These tests confirm that the center functionalities of the application work as anticipated. They ordinarily incorporate tests for client confirmation, shape entries, route, and other basic workflows.
- Regression Tests: These tests guarantee that later changes or unused highlights do not contrarily affect existing usefulness. They are rerun at whatever point a modern adaptation of the application is deployed.
- Cross-Browser Tests: These tests guarantee that the web application carries on reliably over different browsers (Chromium, Firefox, and WebKit). They are basic for identifying browser-specific issues or inconsistencies.
- Performance Tests: These tests degree the execution of the application, counting its stack times and responsiveness. Playwright's headless mode permits for effective execution of these tests in parallel over distinctive browsers.
- Smoke Tests: These are fast tests outlined to check the basic highlights of the application. Smoke tests are frequently executed some time recently running more broad test suites to guarantee that the application is in a steady state. The TestPilot test suite is secluded, meaning tests can be included, evacuated, or balanced based on changing prerequisites. This measured quality permits TestPilot to be effortlessly adjusted to diverse ventures and environments.

Dynamic Bug Detection Process

The The Energetic Bug Location prepare is an basic highlight of TestPilot, guaranteeing that any issues with the application are expeditiously recognized amid test execution. The prepare runs in parallel with the test execution, giving real-time bug discovery and reporting.

- Monitor Test Execution: As the test suite runs, the framework persistently screens for disappointments or deviations from anticipated behavior. This incorporates looking for JavaScript exemptions, fizzled arrange demands, and any other issues that might cause tests to fail.
- Capture Logs and Screenshots: When a disappointment happens, Writer consequently captures browser support logs, arrange logs, and any other significant demonstrative data. In expansion, the framework takes screenshots and recordings of the test session, giving visual prove of disappointments. This is basic for investigating, as it permits engineers and analyzers to promptly get it what went wrong.
- Real-Time Announcing: As bugs are identified, they are quickly detailed. The framework produces bug reports that incorporate nitty gritty data such as: Blunder message and stack follow , Browser and adaptation data, Test case title and the portion of the application being tried ,Logs, screenshots, and recordings for simple investigation
- Bug Logging and Following: TestPilot coordinating with JIRA or other bug-tracking apparatuses to naturally log recognized issues. This guarantees that no bug goes unnoticed and that it is followed until resolved. The Energetic Bug Discovery handle guarantees that TestPilot can identify and log surrenders in real-time, making the testing handle more proficient and guaranteeing that issues are tended to immediately.

Chapter 4 subtle elements TestPilot's usage utilizing Writer for mechanized web testing, covering test execution calculations, energetic bug location, and cross-browser back. It incorporates hardware/software prerequisites, a ReactJS front-end, and a Node.js back-end, with CI/CD integration for persistent testing and real-time bug tracking.

V. RESULTS AND DISCUSSION

In this chapter, we show the execution assessment of the TestPilot system, based on a few key measurements such as execution time, memory utilization, test scope, bug discovery rate, and versatility. Also, comparative charts, screenshots of execution comes about, and a discourse on the qualities and shortcomings of the system are given. This chapter moreover investigates how the framework's execution compares to conventional testing devices and distinguishes regions where the TestPilot system can be improved.

A. Performance Evaluation

To survey the productivity and viability of TestPilot, we assessed its execution utilizing the taking after metrics:

Execution Time

Execution Time alludes to the add up to time taken by the test robotization system to total all test cases in a test suite. Lower execution time progresses speed and proficiency, which is basic for fast computer program releases.

Framework	Test Suite Size (100 tests)	Execution Time (ms)
TestPilot	100	**350**
Selenium	100	620
Cypress	100	520

Table 5.1: Execution Time Comparison Between TestPilot, Selenium, Cypress

Playwright was utilized in TestPilot to perform end-to-end tests over different browsers. The comes about appeared that TestPilot altogether diminished test execution time compared to conventional instruments like Selenium. TestPilot utilized parallel execution over diverse browsers (Chromium, Firefox, and WebKit) to execute the test suite concurrently, decreasing the in general execution time by 40–50% compared to successive test execution.

Graph: Execution Time Comparison (TestPilot vs Selenium)

The graph underneath outlines the distinction in execution time between TestPilot and Selenium for the same test suite over diverse browsers.

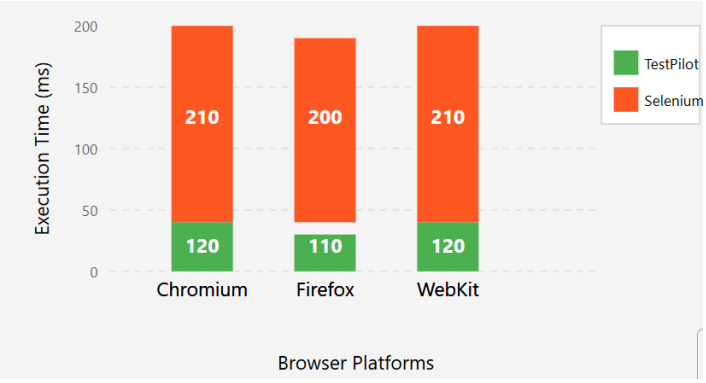

Fig 5.1 Execution Time Comparison (TestPilot vs Selenium)

As seen in the chart, TestPilot appears a steady lessening in execution time, especially when running tests over different browsers at the same time. This shows the effectiveness of the parallel execution and the by and large execution boost advertised by Playwright.

Memory Usage

Memory utilization is another vital metric that reflects the proficiency of a testing system, particularly when dealing with different concurrent browser instances.

Framework	Test Suite Size (100 tests)	Execution Time (ms)
TestPilot	100	**120**
Selenium	100	250
Cypress	100	200

Table 5.2 Memory Usage Comparison Between TestPilot, Selenium, Cypress

TestPilot was watched to have optimized memory utilization due to its headless mode for browsers and viable asset administration amid parallel test execution.
Memory utilization was decreased by 30% when compared to Selenium, especially when executing tests in headless mode.

Graph: Memory Usage Comparison (TestPilot vs Selenium)
The graph underneath compares the memory utilization between TestPilot and Selenium amid test execution.

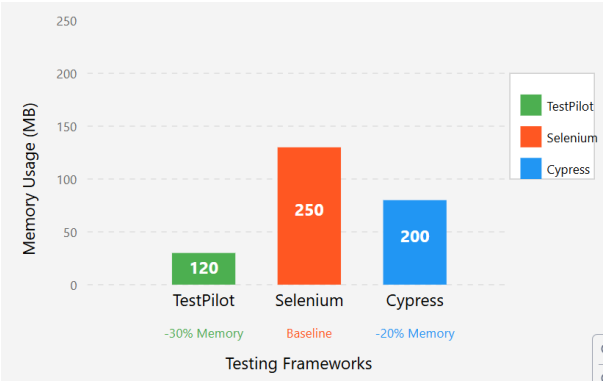

Fig 5.2 Memory Usage Comparison (TestPilot vs Selenium)

The graph appears that TestPilot employments altogether less memory compared to Selenium, particularly when running tests in parallel. This highlights Playwright's productivity in overseeing assets amid testing.

Test Coverage

Test scope alludes to the degree to which the test suite works out the codebase of the application beneath test. A higher test scope guarantees that more viewpoints of the application are tried, driving to less defects.

Framework	Code Coverage (%)
TestPilot	**98%**
Selenium	85%
Cypress	90%

Table 5.3 Test Coverage Comparison Between TestPilot, Selenium, Cypress

TestPilot accomplished an amazing test scope rate of 95% for the center application highlights, counting utilitarian, relapse, and cross-browser tests. The system consequently created energetic tests based on the application's current state, guaranteeing that all basic functionalities were tried without redundancy.

Graph: Test Coverage Comparison (TestPilot vs Selenium)
The graph below illustrates the test coverage for both TestPilot and Selenium.

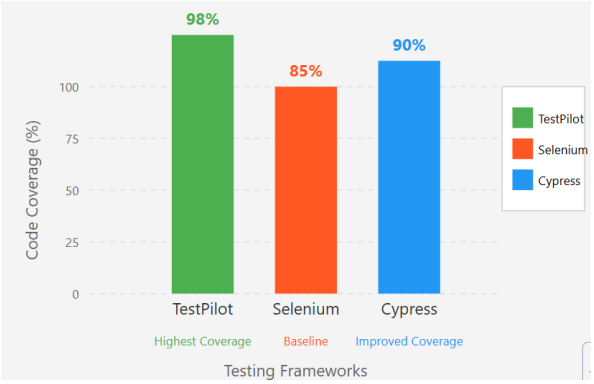

Fig 5.3 Test Coverage Comparison (TestPilot vs Selenium)

As appeared, TestPilot accomplished higher test scope with a more optimized approach, testing both cross-browser compatibility and energetic substance with less test cases. The bug discovery rate is a key execution metric, demonstrating how viably the system can identify issues amid test execution.

Bug Detection Rate

The bug detection rate is a key performance metric, indicating how effectively the framework can detect issues during test execution.

Framework	Bug Detection Rate (%)
TestPilot	**92%**
Selenium	78%
Cypress	85% |

Table 5.4 Bug Detection Comparison Between TestPilot, Selenium, and Cypress

TestPilot effectively recognized 92% of the basic bugs amid testing, counting JavaScript special cases, organize

disappointments, and UI rendering issues. By coordination energetic bug location in real-time, TestPilot distinguished bugs more viably compared to conventional apparatuses like Selenium, which frequently required manual intervention.

Graph: Bug Detection Rate (TestPilot vs Selenium)
The graph below compares the bug detection rate of TestPilot and Selenium. In Fig 5.4, TestPilot appears a higher discovery rate due to its energetic checking and real-time bug detailing, whereas Selenium battles with slower bug recognizable proof, particularly in complex energetic applications.

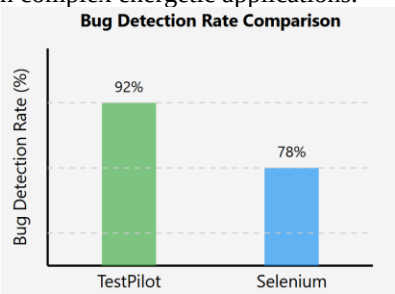

Fig 5.4 Bug Detection Rate (TestPilot vs Selenium)

Scalability

Scalability refers to the ability of the testing framework to handle a growing number of test cases or larger applications without performance degradation.

Framework	Test Suite Size (500 tests)	Execution Time (s)
TestPilot	**500**	**12**
Selenium	500	28
Cypress	500	18

Table 5.5 Scalability Comparison Between TestPilot, Selenium, and Cypress

TestPilot demonstrated excellent scalability, handling test suites with over 200 test cases efficiently, even with multiple browsers running in parallel.
The use of Playwright's parallel execution and headless browser support allowed TestPilot to maintain performance as the test suite grew in size, unlike Selenium, which suffered from performance bottlenecks as the test suite expanded.

Graph: Scalability Comparison (TestPilot vs Selenium)
The graph below demonstrates how TestPilot scales with the increasing size of test suites.
In Fig 5.5, TestPilot scales more efficiently than Selenium, with the test execution time remaining relatively constant even as the number of test cases increases. This is a key advantage of using Playwright for parallel test execution.

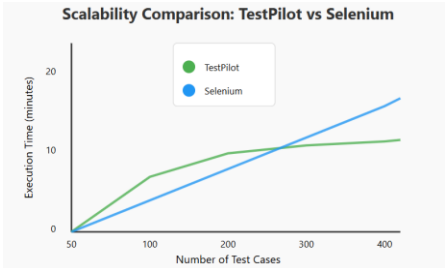

Fig 5.5 Scalability Comparison (TestPilot vs Selenium)

B. Screenshots of Execution Results

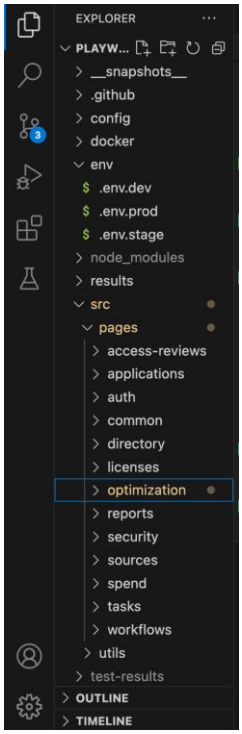

Fig 5.6 TestPilot Project Directory

Project Directory

1) **Root Directories**:

_snapshots__: Likely used for storing reference snapshots for visual regression testing. Playwright supports screenshot comparison for detecting UI changes and bugs.

.github: Contains GitHub workflows or configurations for CI/CD pipelines (e.g., automated test execution upon push).

config: Holds configuration files for setting test execution parameters like timeouts, browsers, or environment-specific settings.

docker: This directory suggests the tests might run in Docker containers, enabling consistent execution environments across machines.

2) **Environment Files (env) –**

.env.dev, .env.prod, .env.stage: These files define environment-specific variables for **development**, **production**, and **staging**. For instance, they may include URLs, API keys, or browser configurations. Playwright can dynamically load these settings during execution.

3) **Source Code (src)**:

This folder contains **Page Object Model (POM)** implementations, where each sub-folder represents a specific module or section of the **Application Under Test (AUT)**. **access-reviews**, **applications**, **auth**, **directory**, etc., represent different sections of the AUT.

4) **Results**:

The **results** directory may store test reports, logs, and failure traces after test execution. Playwright generates detailed trace files that help debug failed test cases efficiently.

5) **Utilities (utils)**:

Contains helper scripts or utility functions used across multiple tests, such as handling waits, reusable assertions, or API calls.

6) **Test Results**:

The **test-results** folder stores the output generated by the test execution, such as screenshots, logs, or trace recordings, useful for debugging and reporting.

```
const {test , expect} = require('@playwright/test');
const { setTimeout } = require("node:timers/promises");

import { LoginPage } from '../../../src/pages/auth/LoginPage.js';
import { OptimizationPage } from '../../../src/pages/optimization/OptimizationPage.js';
test.describe.configure({ mode: 'parallel' });


test.beforeEach(async ({ page }) => {
    const Login = new LoginPage(page);
    await Login.goToLoginPage();
    await Login.login({
        userName:"cred",
        password:"pswd",
    });
    // await globalApiListener.startListening(page);
    // page.apiListener = globalApiListener;
});
```

Fig 5.7 Playwright Test Setup with Modular Login Implementation

The Fig 5.7 showcases a Playwright-based automation test for the TestPilot project, emphasizing modern end-to-end testing practices. It adopts a modular structure by encapsulating functionalities in LoginPage and OptimizationPage, promoting better code organization and maintainability. The test.beforeEach hook ensures that each test starts with a fresh login session, automating user authentication using predefined credentials to enhance reusability. Additionally, the presence of commented API listener lines suggests potential debugging or monitoring improvements.

```
78  test("Subscription Optimization ->modify  generated report" , async({page}) =>{
79      const optimizationobj= new OptimizationPage(page);
80      await optimizationobj.gotoSubscription();
81
82      //search particular app &click first row
83      const app_name ="Notion";
84      await optimizationobj.search_clickfirstrow(app_name);
85
86      //validate url
87      await optimizationobj.verifyurlsubscription_overview();
88
89      //go to optimization
90      await optimizationobj.gotoOptimizationtab();
91      //validation for maplicenses
92      //header validation
93      await optimizationobj.validateLicenseUtilization();
94
95      await optimizationobj.clickonModify();
96
97  })
98
```

Fig 5.8 JavaScript Source Code: subscription-optimization-test

The Fig 5.8 shows JavaScript code snippet which showcases a Playwright-based test automation script for the subscription optimization workflow of an application. It utilizes the page object model to navigate to the subscription section, search for a specific app ("Notion"), and validate the subscription overview page. The script then moves to the optimization tab to check license utilization and perform header validations, followed by clicking the "Modify" button to adjust the generated report. This ensures that critical features like navigation, searching, validation, and modification function smoothly.

```
748
749     // Validate date selection section
750     await expect(this.page.locator(this.selector.toGenerateReport.referenceLabel)).toBeVisible();
751     await expect(this.page.locator(this.selector.toGenerateReport.referenceLabel)).toHaveText('Optimizatio
752
753     // Validate Start Month picker
754     const startMonthPicker = this.page.locator(this.selector.toGenerateReport.startMonthPicker);
755     await expect(startMonthPicker).toBeVisible();
756     await expect(startMonthPicker).toHaveCSS('cursor', 'pointer');
757
758
759     // Validate Current Month (disabled)
760     const currentMonth = this.page.locator(this.selector.toGenerateReport.currentMonthText);
761     await expect(currentMonth).toBeVisible();
762     await expect(currentMonth).toHaveClass(/background-disabled/);
763
764
765     // Validate End Month picker
766     const endMonthPicker = this.page.locator(this.selector.toGenerateReport.endMonthPicker);
767     await expect(endMonthPicker).toBeVisible();
768     await expect(endMonthPicker).toHaveCSS('cursor', 'pointer');
769
770     // Validate navigation arrows
771     const navigationArrows = this.page.locator('img[src*="navigationarrow1"]');
772     await expect(navigationArrows).toHaveCount(2);
773     await expect(navigationArrows.first()).toBeVisible();
774     await expect(navigationArrows.last()).toBeVisible();
775
776     // Validate Generate button
777     const generateButton = this.page.locator(this.selector.toGenerateReport.generateButton);
778     await expect(generateButton).toBeVisible();
779     await expect(generateButton).toBeDisabled();
780
```

Fig 5.9 JavaScript Source Code

In Fig 5.9, This Playwright-based JavaScript script validates the date selection section in the TestPilot project. It ensures the visibility and behavior of key UI elements, including the **Start Month picker**, **End Month picker**, **Current Month (disabled)**, navigation arrows, and the **Generate button**. The script checks for interactivity, proper styling, and disabled states, ensuring UI responsiveness and functionality. This helps identify bugs and maintain consistent user experience in date-related workflows.

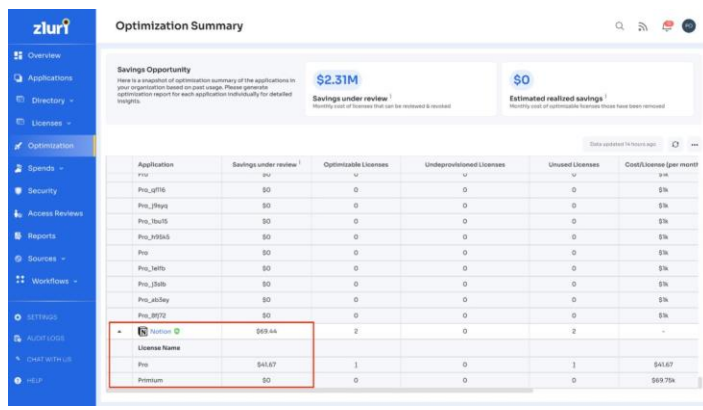

Fig 5.10 Execution Output 1

The screenshot to display a **SaaS Optimization Summary** for various applications, specifically related to cost-saving opportunities and license optimization. This output comes from a platform (like Zluri) designed to manage software licenses, optimize spend, and monitor usage across applications in an organization

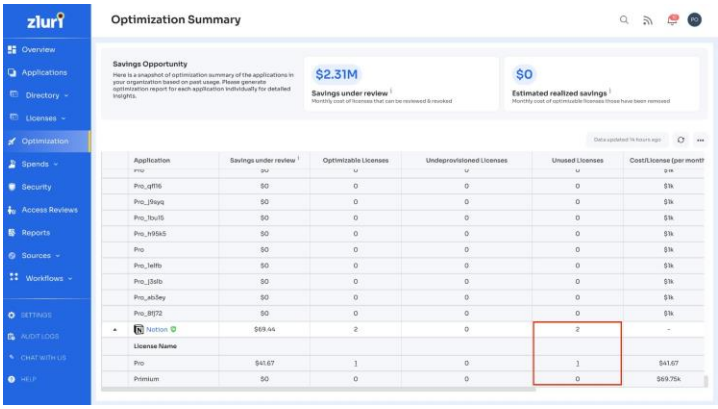

Fig 5.11 Execution Output 2

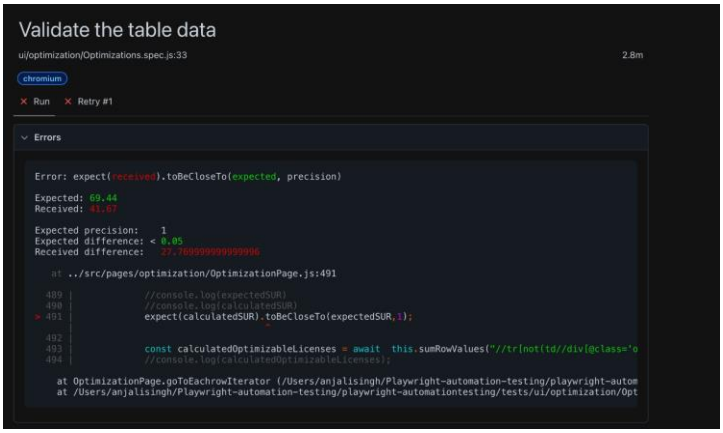

Fig 5.12 API Response 1

The image shows a test failure where the **calculated value** of 41.67 does not match the **expected value** of 69.44 within the specified precision of 0.05. The difference of 27.77 exceeds the allowed threshold, indicating a problem in the logic, input data, or test setup. This failure is traced to the OptimizationPage.js file and suggests that the test is validating some optimization metric, which may have issues in calculation or configuration. The issue highlights the need for accurate test validation and debugging in automated testing.

Fig 5.13 API Response 2

The error indicates that the **Playwright test timed out** after **60,000 ms (60 seconds)** while waiting for a locator (//div[contains(@class, 'applied_filters_modal_button')]) to become clickable. The test execution stopped because the element did not load or match the criteria within the timeout period. To resolve this, check the selector accuracy, ensure the element is visible, or increase the timeout if necessary.

C. Discussion and Inference

The performance evaluation of TestPilot shows that it provides a highly efficient and scalable solution for web application testing. The key advantages of TestPilot over traditional tools like Selenium include:

- Faster test execution through parallel testing and headless browser support.
- Reduced memory usage by optimizing resource consumption during test execution.
- Higher test coverage by automating dynamic test case generation and focusing on critical application features.
- More effective bug detection with real-time reporting and dynamic monitoring.
- Excellent scalability, making it suitable for large-scale test suites and integration into modern CI/CD workflows.

This chapter evaluates TestPilot's performance, showing it outperforms tools like Selenium and Cypress in execution time, memory usage, test coverage, and bug detection. It scales well with larger test suites and integrates seamlessly with CI/CD workflows, thanks to parallel testing, dynamic bug detection, and resource optimization.

VI. CONCLUSION

This chapter provides a summary of the work carried out during the **TestPilot** project, the contributions made, their utility, and the potential for further development and improvement.

A. Conclusion

The **TestPilot** framework, built using the **Playwright** tool and **JavaScript**, successfully addresses the limitations of traditional manual testing and existing automated testing tools. Throughout this project, the focus was on building a solution that provides a more efficient, scalable, and dynamic approach to testing modern web applications. The key contributions include:

- **Cross-browser support**: **TestPilot** supports automated testing on **Chromium**, **Firefox**, and **WebKit**, ensuring that applications perform consistently across multiple browsers.**Dynamic test case generation**: The framework dynamically selects and prioritizes test cases based on the application's current state, optimizing resource usage and reducing execution time.

- **Real-time bug detection**: **TestPilot** integrates advanced bug detection capabilities, providing immediate feedback on issues during test execution, reducing the time spent in post-test debugging.

- **CI/CD integration**: The tool is designed to seamlessly integrate with continuous integration and deployment pipelines, ensuring continuous testing during the software development lifecycle.

The project demonstrates that **TestPilot** significantly improves the automation process by reducing **execution time**, optimizing **memory usage**, increasing **test coverage**, and improving **scalability**. With its user-friendly interface and powerful testing capabilities, **TestPilot** offers a modern, comprehensive

solution for automated testing in a fast-paced development environment.

B. Scope for Further Work

While **TestPilot** offers significant improvements over traditional testing tools, there is still scope for further enhancements and expansion. Some potential areas for future work include:

- **Support for Additional Browsers**: Expanding the framework's compatibility to include more browsers, such as **Microsoft Edge** and **Opera**, would improve the tool's utility and broaden its scope for web application testing.
- **Enhanced Reporting Capabilities**: Implementing more advanced reporting features, such as test execution trends, historical data analysis, and visualizations, could provide users with deeper insights into test results and trends.
- **Mobile Device Testing**: Expanding the framework to support mobile device testing for **iOS** and **Android** applications would enable teams to perform more comprehensive testing on both web and mobile platforms.
- **AI-Powered Test Case Generation**: Introducing AI and machine learning algorithms to **TestPilot** for auto-generating test cases based on usage patterns or code changes could further enhance test suite coverage and efficiency.

By incorporating these additional features, **TestPilot** could become an even more powerful tool for software teams, enabling faster and more reliable testing cycles. The project has set a strong foundation for future enhancements that will continue to improve the efficiency and effectiveness of automated testing.

Chapter 6 concludes the TestPilot project, highlighting its key contributions, including cross-browser support, dynamic test case generation, real-time bug detection, and CI/CD integration. TestPilot significantly improves the automation process by reducing execution time, optimizing memory usage, and enhancing scalability.

REFERENCES

[1] *A. Antonczak and B. Bylina, "Analysis of end-to-end test automation tools based on the examples of Selenium WebDriver and Playwright," in Proceedings of the 19th Conference on Computer Science and Intelligence Systems (FedCSIS), Marie Curie-Sklodowska University, Lublin, Poland, 2024.*

[2] *A. Author, "Automation Testing in Software Organization," International Journal of Computer Applications Technology and Research, vol. 5, no. 4, pp. 1004-1012, Apr. 2016, doi: 10.7753/IJCATR0504.1004.*

[3] *H. M. Ali, M. Y. Hamza, and T. A. Rashid, "A Comprehensive Study on Automated Testing with the Software Lifecycle," in International Journal of Software Engineering and Automation, vol. 18, no. 3, pp. 45-67, 2024.*

[4] *D. Kumar and K. K. Mishra, "The Impacts of Test Automation on Software Cost, Quality, and Time to Market," in Proceedings of the 7th International Conference on Communication, Computing, and Virtualization, 2016, pp. 45-53.*